Aaradhya Dev Tamrakar

AI/ML Engineering | Applied ML | AI Workflows | Embedded AI

Kathmandu, Nepal | aaradhyadevtmr@gmail.com | +977 9844602050
aaradhyadt.github.io | github.com/AaradhyaDT | LinkedIn: aaradhya-dev-tamrakar

---

Final-year electronics engineering student building applied ML systems and AI workflow tools. Uses AI for development and turns recurring work into reusable automation and deterministic checks. Work spans demographic performance auditing, multi-provider coordination, and embedded ML software. Seeking AI/ML engineering opportunities.

Education

Bachelor of Engineering in Electronics, Communication and Information Engineering

Kathmandu Engineering College, IOE, Tribhuvan University | Expected January 2027

- Final year. Coursework includes machine learning, signals and systems, control systems, and digital electronics.

Selected Engineering Projects

BiasAperture | Demographic performance auditing

Python | AIF360 | Fairlearn | Statistical testing | Jinja2

- Co-developed a prediction-auditing and reporting pipeline as a two-person Fuse AI Fellowship capstone.
- Focus: canonical binary group-rate metrics, cohort support, and traceable outputs. Statistical validation and matched model comparisons remain active work.

Fleet Orchestrator | AI workflow coordination

Python | FastAPI | Provider adapters | MCP

- Built coordination tooling for multi-step AI workflows, with task state, worker adapters, scheduling, and status inspection.
- Uses reusable interfaces and local tooling to reduce repeated orchestration work; runtime availability depends on providers and quotas.

SPARK | Wearable fall-detection project

ESP32-S3 | TFLite Micro | Python | Gateway APIs

- Project lead for ML modelling, gateway processing, and explainability in a four-person team; firmware and hardware have separate owners.
- Offline model and software artifacts are available. Hardware, on-device, BLE, and volunteer cohort validation remain pending.

Agent-Customization-Sync | Cross-IDE configuration tooling

Python | SHA-256 | MCP configuration

- Built configuration snapshots, guarded restore, rules transpilation, and drift audits across development environments.
- Audits instruction overhead and links shared skills to reduce configuration duplication.

Aaradhya Dev Tamrakar

AI/ML Engineering | Experience and technical background

Applied ML Coursework

ML and agent MLOps workflows

MLflow | FastAPI | Evidently | Airflow

Built fellowship workflows for experiment tracking, serving, drift monitoring, and scheduled regression evaluation. Agent evaluation tracks prompts and traces; judge-model scores need separate calibration.

Trustworthy AI guardrails

Python | Input and output validation | Pydantic

Implemented coursework checks for prompt injection, sensitive information, output schema compliance, and groundedness, with a golden-case evaluation set.

Fellowships

Fuse AI Fellow | Fusemachines

2026 cohort

- Built applied ML and agentic coursework spanning classification, explainability, APIs, and MLOps; co-developed the BiasAperture capstone with Tisha Manandhar.

NSSR DataCamp Fellow | Cohort 2

2026

- Completed applied GenAI projects; continuing SQL and Python data-analysis pathways.

Technical Skills

ML and evaluation

Python, scikit-learn, Pandas, NumPy, XGBoost, SHAP, AIF360, Fairlearn

AI systems and backend

FastAPI, Docker, SQL, PostgreSQL, SQLite, asyncio, MCP

Embedded and vision

C/C++, ESP32-S3, TFLite Micro, Arduino, MediaPipe, Kotlin

Engineering practice

Git, automated tests, reproducible configurations, model evaluation

Leadership and Engineering Service

IEEE KEC KTM Student Branch | Vice Chair

2026-Present

- Coordinate branch planning, technical activities, and committee communication; previously Vice Secretary, 2025-2026.

KEC Makerspace | Ambassador

June 2026-Present

- Maintain website content and interface fixes.

Electronics Project Club | Event Manager

2026-Present

- Coordinate event logistics and delivery; previously Resource Manager, 2024-2026.